

POSITIVE EIGENVALUES OF THE RELATIVE SU(2) BFSS HAMILTONIAN

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ABSTRACT. We prove that the relative SU(2) BFSS Hamiltonian has infinitely many positive eigenvalues tending to infinity, with square-integrable eigenvectors. The operator is defined by closing the gauge-invariant supercharge form. This refutes, at $N = 2$, the exclusion of normalizable positive-energy states stated in the original BFSS paper.

1. INTRODUCTION

The Banks–Fischler–Shenker–Susskind (BFSS) model is supersymmetric matrix quantum mechanics obtained by reducing ten-dimensional supersymmetric Yang–Mills theory to one time dimension. Its quartic potential vanishes on commuting matrices, leaving unbounded directions of classical escape. For the relative SU(2) Hamiltonian defined below, we prove an unbounded sequence of positive eigenvalues with square-integrable eigenvectors. The proof selects Spin(9) sectors in which symmetry excludes low transverse oscillator levels, so the remaining oscillator energy confines those sectors. We first specify the operator and its physical Hilbert space.

We study the relative model, with the free center of mass removed, and use the following normalization. The color basis $T_a = \sigma_a/\sqrt{2}$, where σ_a are the Pauli matrices, satisfies

$$\mathrm{Tr}(T_a T_b) = \delta_{ab}, \quad [T_a, T_b] = i f_{abc} T_c, \quad f_{abc} = \sqrt{2} \varepsilon_{abc}.$$

Write the nine traceless bosonic matrices as $X_i = \sum_{a=1}^3 x_i^a T_a$ and group their coefficients into

$$x = (x^1, x^2, x^3) \in (\mathbb{R}^9)^3, \quad x^a = (x_i^a)_{i=1}^9, \quad |x|^2 = \sum_{a=1}^3 |x^a|^2.$$

An orthogonal change of color basis acts unitarily on the bosonic and fermionic variables, so this choice represents the SU(2) Hamiltonian in the stated normalization.

Let $\mathcal{F}$ be an irreducible complex Clifford module with self-adjoint generators θ_α^a , $1 \leq \alpha \leq 16$ and $1 \leq a \leq 3$, such that

$$\{\theta_\alpha^a, \theta_\beta^b\} = \delta_{\alpha\beta} \delta_{ab}.$$

In particular, $\mathcal{F}$ is finite dimensional. Fix real symmetric matrices $\gamma^1, \dots, \gamma^9$ of size 16 with

$$\{\gamma^i, \gamma^j\} = 2\delta_{ij} I, \quad \gamma^{ij} = \frac{1}{2} [\gamma^i, \gamma^j].$$

For $h \in \mathrm{SU}(2)$, let $O_h \in \mathrm{SO}(3)$ be its adjoint action in the basis T_a . Its action on the real generator label space $\mathbb{R}^3 \otimes \mathbb{R}^{16}$ is $O_h \otimes I_{16} \in \mathrm{SO}(48)$. Since SU(2) is simply connected, this homomorphism lifts uniquely to Spin(48). Let V_h be the unitary Clifford action of that lift on $\mathcal{F}$, using the normalized generators $\sqrt{2} \theta_\alpha^a$. The gauge action on functions is

$$(\mathcal{G}_h \Psi)(x) = V_h \Psi(O_h^{-1} x),$$

where O_h^{-1} acts on the color index of x . Define

$$\mathcal{H} = [L^2(\mathbb{R}^{27}; \mathcal{F})]^{\mathrm{SU}(2)}, \quad \mathcal{C} = [C_c^\infty(\mathbb{R}^{27}; \mathcal{F})]^{\mathrm{SU}(2)}.$$

On $\mathcal{C}$, put

$$(1.1) \quad Q_\alpha = -i\gamma_{\alpha\beta}^i \theta_\beta^a \partial_{ia} + \frac{1}{2} f_{abc} x_i^b x_j^c \gamma_{\alpha\beta}^{ij} \theta_\beta^a, \quad q(\Psi) = \frac{1}{16} \sum_{\alpha=1}^{16} \|Q_\alpha \Psi\|_2^2.$$

Here $\partial_{ia} = \partial/\partial x_i^a$, and repeated indices are summed. The tensors δ_{ab} and f_{abc} are invariant under O_h , so $\mathcal{G}_h Q_\alpha \mathcal{G}_h^{-1} = Q_\alpha$ on $C_c^\infty(\mathbb{R}^{27}; \mathcal{F})$. Each Q_α is symmetric on $\mathcal{C}$. Gauge averaging shows that $\mathcal{C}$ is dense in $\mathcal{H}$. Consequently the operator $\Psi \mapsto (Q_1 \Psi, \dots, Q_{16} \Psi)$ is closable: convergence of Ψ_n to zero and of each $Q_\alpha \Psi_n$ forces every limit to vanish by symmetry. Thus q is closable. Let H be the nonnegative self-adjoint operator associated with its closure, still denoted by q . This is the relative $SU(2)$ BFSS Hamiltonian considered here.

Theorem 1.1. *There are an orthonormal sequence $\Psi_j \in \text{Dom}(H)$ and real numbers $E_j > 0$ tending to infinity such that*

$$H\Psi_j = E_j \Psi_j.$$

In particular, $\sigma_p(H) \cap (0, \infty)$ is infinite.

The mechanism is visible in the quartic bosonic potential

$$V(x) = \sum_{a < b} |x^a \wedge x^b|^2.$$

It vanishes when the three spatial vectors are collinear, leaving an unbounded set along which confinement can fail. Fixing one nonzero vector $u = x^a$ separates each other vector into a component parallel to u and a component in $u^\perp$. The potential together with the transverse kinetic energy bounds a harmonic oscillator in these sixteen coordinates, with frequency proportional to $|u|$.

The symmetry determines which oscillator levels are available. The stabilizer of u in $\text{Spin}(9)$ is $\text{Spin}(8)$. If a $\text{Spin}(9)$ representation has highest weight $\lambda = (\lambda_1, \lambda_2, \lambda_3, \lambda_4)$, every constituent of its restriction to this stabilizer has highest weight ν with $\nu_1 \geq \lambda_2$. If n is the oscillator level and M is a fixed bound for the contribution from the fermions, a stabilizer type ν shared by this restriction and level n tensored with the fermions must satisfy

$$\lambda_2 \leq \nu_1 \leq n + M.$$

A large λ_2 therefore removes low levels. This is why the second coordinate matters: the first coordinate alone does not impose this restriction on stabilizer types.

The resulting oscillator energy grows linearly in $|u|$. Squaring the supercharges also produces a fermionic multiplication term that is linear in x and may be negative. By excluding sufficiently many oscillator levels, the positive linear coefficient dominates this term. Averaging the three choices of color then gives a form bound that confines the full variable x within the chosen rotation sector. For every sector above a fixed threshold for λ_2 , the restricted Hamiltonian consequently has compact resolvent and trivial kernel. Explicit gauge-invariant polynomials show that infinitely many such physical sectors are infinite dimensional; any one of them supplies the unbounded sequence of positive eigenvalues.

The spectral history makes the distinction between the spectral set and normalizable eigenstates essential. For the finite matrix Hamiltonians they studied, de Wit, Lüscher, and Nicolai proved that the spectral set is $[0, \infty)$ and explicitly left open the possibility of normalizable eigenstates within that interval [6, Section 1]. In their discussion of relative motion after removing the free center of mass, Banks, Fischler, Shenker, and Susskind stated that, besides the zero-energy threshold states, “No other normalizable bound states can occur” [2, Section 5, after Equation (5.4)]. Theorem 1.1 refutes this exclusion for the exact relative operator defined here at $N = 2$; the large- N Matrix Theory conjecture is a separate question.

Positive-energy binding had appeared in a qualified form in the work of Danielsson, Ferretti, and Sundborg. They obtained normalizable excited wavefunctions in a Born–Oppenheimer approximation to the same relative SU(2) model, anticipated decay beyond the approximation, and described the states as metastable [4, Sections 3.3 and 4–5]. Smilga later conjectured families of normalized excited states in a large- N thermodynamic analysis [10, Section 3]. Sethi and Stern’s rotational invariance theorem concerns normalizable zero-energy states [8, Sections 3.2 and 4.2], whereas the eigenvalues proved to exist here are positive.

Transverse quantization has classical scalar precedents in Simon’s x^2y^2 oscillator argument [9, Section 2] and Lüscher’s estimates for bosonic Yang–Mills Hamiltonians [7, Appendix D]. Davies and Simon used parity to remove the transverse constant mode of a symmetric Neumann horn and obtain a reducing sector with compact resolvent [5, Theorem 3.1 and Step 7]. In a supersymmetric matrix model with two spatial matrices and SU(2) $\times$ SO(2) symmetry, Aref’eva, Koshelev, and Medvedev decomposed the problem into angular sectors and used a Born–Oppenheimer analysis to argue for sectors with purely discrete spectrum, deriving asymptotic spectral formulas [1, Sections 5–6]. Here the decisive model-specific step is to use orthogonal branching and the finite fermionic weight set to exclude any prescribed finite number of transverse oscillator levels. More generally, restriction to the stabilizer of an escape direction can identify symmetry sectors in which transverse excitation restores confinement.

2. THE QUADRATIC FORM AND ROTATIONS

We first isolate the scalar potential and the fermionic term. We then show that rotation types reduce the closed form and its associated operator. All differential computations in this section take place on compactly supported smooth functions.

Lemma 2.1. *For $\Psi \in \mathcal{C}$,*

$$(2.1) \quad q(\Psi) = \frac{1}{2} \|\nabla \Psi\|_2^2 + \int_{\mathbb{R}^{27}} V(x) \|\Psi(x)\|^2 dx + \int_{\mathbb{R}^{27}} \langle \Psi(x), B(x) \Psi(x) \rangle dx,$$

where

$$(2.2) \quad V(x) = \sum_{a < b} |x^a \wedge x^b|^2, \quad B(x) = \frac{i}{2} f_{abc} x_i^a \gamma_{\beta\delta}^i \theta_{\beta}^b \theta_{\delta}^c.$$

The matrix $B(x)$ is Hermitian. There is a constant $C < \infty$ such that

$$(2.3) \quad \|B(x)\|_{\text{op}} \leq C|x| \quad (x \in \mathbb{R}^{27}).$$

Proof. The calculation also holds on $C_c^\infty(\mathbb{R}^{27}; \mathcal{F})$ before taking gauge invariants. Put

$$y_{ij}^a = f_{abc} x_i^b x_j^c, \quad A_\alpha = -i \gamma_{\alpha\beta}^i \theta_\beta^a \partial_{ia}, \quad W_\alpha = \sum_{j < k, c} y_{jk}^c \gamma_{\alpha\delta}^{jk} \theta_\delta^c.$$

The antisymmetry in j, k makes $Q_\alpha = A_\alpha + W_\alpha$. The gamma relations give

$$\text{tr}((\gamma^i)^T \gamma^j) = 16\delta_{ij}, \quad \text{tr}((\gamma^{ij})^T \gamma^{kl}) = 16\delta_{ik}\delta_{jl} \quad (i < j, k < l).$$

Indeed, each indicated matrix has squared Hilbert–Schmidt norm 16, and a product of two or four distinct gamma matrices has trace zero: conjugation by one of its factors changes its sign. Clifford anticommutation, followed by these identities, gives on the smooth core

$$\frac{1}{16} \sum_{\alpha} A_\alpha^2 = -\frac{1}{2} \Delta, \quad \frac{1}{16} \sum_{\alpha} W_\alpha^2 = \frac{1}{2} \sum_{a, j < k} (y_{jk}^a)^2 I.$$

For the latter identity, the coefficients y_{jk}^a commute, so the two orders of each Clifford product combine into an anticommutator. Since $f_{abc} = \sqrt{2}\varepsilon_{abc}$, the last scalar is $\sum_{a < b} |x^a \wedge x^b|^2$.

In the cross term, the first-order part is

$$-\frac{i}{16} \sum_{i,a,j < k} y_{jk}^a \operatorname{tr}((\gamma^i)^T \gamma^{jk}) \partial_{ia} = 0,$$

because γ^i is symmetric and γ^{jk} is skew-symmetric. The remaining multiplication operator is

$$-\frac{i}{16} \sum_{\substack{i,a,\beta,\delta \\ j < k,c}} (\partial_{ia} y_{jk}^c) (\gamma^i \gamma^{jk})_{\beta\delta} \theta_\beta^a \theta_\delta^c.$$

Here

$$\partial_{ia} y_{jk}^c = f_{cab} \delta_{ij} x_k^b + f_{cba} x_j^b \delta_{ik}, \quad \gamma^j \gamma^{jk} = \gamma^k, \quad \gamma^k \gamma^{jk} = -\gamma^j.$$

Each spatial index occurs in eight pairs $j < k$. Substitution gives

$$-\frac{i}{2} f_{cab} x_i^b \gamma_{\beta\delta}^i \theta_\beta^a \theta_\delta^c = \frac{i}{2} f_{abc} x_i^a \gamma_{\beta\delta}^i \theta_\beta^b \theta_\delta^c,$$

where the last equality uses antisymmetry of f and relabels the indices. Integration by parts now proves (2.1) on $\mathcal{C}$.

Whenever $f_{abc} \neq 0$, the generators θ_β^b and θ_δ^c anticommute. Their product is anti-Hermitian, so the displayed B is Hermitian. It is a linear function of x with coefficients in the finite-dimensional space $\operatorname{End}(\mathcal{F})$. This proves (2.3). $\square$

Let $G = \operatorname{Spin}(9)$, and write $R_g \in \operatorname{SO}(9)$ for the vector representation. Clifford multiplication gives a real orthogonal spin representation S_g on $\mathbb{R}^{16}$ satisfying

$$(2.4) \quad S_g \gamma(v) S_g^{-1} = \gamma(R_g v), \quad \gamma(v) = v_i \gamma^i.$$

For $w \in \mathbb{R}^{16}$, write $\theta^a(w) = w_\alpha \theta_\alpha^a$.

Lemma 2.2. *There is a unitary representation U of G on $\mathcal{F}$ such that*

$$U_g \theta^a(w) U_g^{-1} = \theta^a(S_g w).$$

It commutes with the gauge action. The representation

$$(2.5) \quad (D_g \Psi)(x) = U_g \Psi(R_g^{-1} x)$$

on $\mathcal{H}$ preserves $\mathcal{C}$ and the closed form q .

For an irreducible G representation of type λ , let $\mathcal{H}_\lambda$ be its isotypic subspace and P_λ the orthogonal projection onto that subspace. Then $P_\lambda \mathcal{C}$ is a form core for q on $\mathcal{H}_\lambda$, and $\mathcal{H}_\lambda$ reduces H . The operator associated with the restricted form is the part $H_\lambda = H|_{\operatorname{Dom}(H) \cap \mathcal{H}_\lambda}$.

Proof. The action S_g on each of the three color copies gives a homomorphism $G \rightarrow \operatorname{SO}(48)$. Since G is simply connected, it lifts to $\operatorname{Spin}(48)$. Acting on $\mathcal{F}$ by the normalized Clifford generators $\sqrt{2} \theta_\alpha^a$ gives U with the stated property.

The actions of G and $\operatorname{SU}(2)$ on the generator labels commute. Their unitary commutator on $\mathcal{F}$ therefore commutes with every Clifford generator and is scalar by irreducibility. For fixed $g \in G$ this scalar, as a function of the gauge transformation, is a continuous character of $\operatorname{SU}(2)$. Such a character is trivial. Thus the actions commute on $\mathcal{F}$, and (2.5) preserves $\mathcal{H}$.

The pullback in (2.5) sends the coordinate and derivative vectors to their R_g^{-1} transforms. The column of Clifford generators transforms by S_g^{-1} . Applying (2.4), and its consequence for γ^{ij} , to the two terms in (1.1) gives

$$D_g Q_\alpha D_g^{-1} = (S_g^{-1})_{\alpha\beta} Q_\beta \quad \text{on } \mathcal{C}.$$

The orthogonality of S_g proves $q(D_g \Psi) = q(\Psi)$. Rotations preserve smoothness, and the rotated supports of a compact set lie in a common ball, so $D_g \mathcal{C} = \mathcal{C}$.

The action is strongly continuous for the form norm $\|\Psi\|_q^2 = \|\Psi\|_2^2 + q(\Psi)$. On $\mathcal{C}$ this follows directly from smoothness, (1.1), and compact support; it extends to the closure because every D_g is a form-norm isometry. For normalized Haar measure and the irreducible character χ_λ ,

$$P_\lambda = (\dim V_\lambda) \int_G \overline{\chi_\lambda(g)} D_g dg.$$

This integral is bounded in form norm. Differentiation under the integral and the common support bound show that it maps $\mathcal{C}$ into $\mathcal{C}$. If $\Psi \in \text{Dom}(q) \cap \mathcal{H}_\lambda$, project any core approximation to Ψ to obtain a form-norm approximation in $P_\lambda \mathcal{C}$. This proves the core assertion.

Invariance of the closed form implies that its resolvent commutes with every D_g : this follows, for example, from uniqueness of the solution of

$$q(u, v) + \langle u, v \rangle = \langle f, v \rangle \quad (v \in \text{Dom}(q))$$

defining $u = (H + 1)^{-1}f$, where $q(u, v)$ denotes the polarization of q . The resolvent therefore commutes with P_λ . This proves reduction and identifies the restricted operator with the stated part of H . $\square$

3. TRANSVERSE CONFINEMENT IN LARGE ROTATION TYPES

Fix a color a , a vector $u \in \mathbb{R}^9 \setminus \{0\}$, and two real numbers s_b for $b \neq a$. Put $\hat{u} = u/|u|$. For $z = (z_b)_{b \neq a} \in (u^\perp)^2$, define $x(u, s, z)$ by its color components:

$$(3.1) \quad x(u, s, z)^a = u, \quad x(u, s, z)^b = s_b \hat{u} + z_b \quad (b \neq a).$$

The transverse space $(u^\perp)^2$ has dimension sixteen. On this slice, the potential satisfies

$$(3.2) \quad V(x(u, s, z)) \geq \sum_{b \neq a} |u \wedge x(u, s, z)^b|^2 = |u|^2 |z|^2, \quad |z|^2 = \sum_{b \neq a} |z_b|^2.$$

The subgroup $K_u \subset G = \text{Spin}(9)$ fixing u acts on the transverse pair and is conjugate to $\text{Spin}(8)$. We will compare the K_u types on a slice with those in each transverse oscillator level. The needed lower bound on slice weights comes from restricting a G type to this stabilizer.

First use the reference stabilizer $K = \text{Stab}_G(e_9) \simeq \text{Spin}(8)$. Choose weight coordinates $\epsilon_1, \dots, \epsilon_4$ for rotations in four orthogonal planes in $\mathbb{R}^8 = e_9^\perp$. The positive roots of K are $\epsilon_j \pm \epsilon_l$ for $j < l$; those of G include these and the ϵ_j . Dominant weights have the forms

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4 \geq 0 \quad (G), \quad \nu_1 \geq \nu_2 \geq \nu_3 \geq |\nu_4| \quad (K),$$

with coordinates all integral or all half-integral in each weight.

Lemma 3.1 (Orthogonal restriction). *If the K type of highest weight ν occurs in the restriction of the G type of highest weight λ , then*

$$(3.3) \quad \nu_1 \geq \lambda_2.$$

Proof. This is the part of the classical orthogonal interlacing rule needed here; compare [3, Proposition 3.7]. We give a character proof that includes the half-integral cases.

Use formal torus monomials Z^w , allowing half-integral exponents, and write Δ_G, Δ_K for the Weyl denominators. The half-sums of positive roots are

$$\rho_K = (3, 2, 1, 0), \quad \rho_G = \rho_K + \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right).$$

Thus $\Delta_G = \Delta_K \prod_i (Z_i^{1/2} - Z_i^{-1/2})$. Set $L_j = \lambda_j + 4 - j$. The Weyl character formula for type B_4 gives

$$(3.4) \quad \chi_\lambda \Delta_K = \det \left(\frac{Z_i^{L_j+1/2} - Z_i^{-L_j-1/2}}{Z_i^{1/2} - Z_i^{-1/2}} \right)_{i,j=1}^4 = \det \left(\sum_{s=-L_j}^{L_j} Z_i^s \right)_{i,j=1}^4,$$

where s increases in steps of one.

Let m_η be the multiplicity of the K type η in the restriction, and let W_K be the Weyl group of K . The Weyl character formula for K gives the second expression

$$\chi_\lambda \Delta_K = \sum_{\eta} m_{\eta} \sum_{w \in W_K} \det(w) Z^{w(\eta + \rho_K)}.$$

Put $t = \nu + \rho_K$. Its coordinates satisfy $t_1 > t_2 > t_3 > |t_4|$, so t is regular dominant for type D_4 , including when $t_4 = 0$. The coefficient of Z^t in this expression is therefore m_ν : only $\eta = \nu$ and the identity Weyl element contribute. In the determinant in (3.4), row i involves only Z_i , so coefficient extraction acts row by row. If the coordinate lattices of λ and ν differ, the coefficient is zero. Otherwise it is

$$\det(\mathbf{1}_{\{|t_i| \leq L_j\}})_{i,j=1}^4.$$

The L_j strictly decrease, and so do the $|t_i|$. Each row is an initial string of ones followed by zeros, and the string lengths are nondecreasing down the rows. A nonzero determinant has neither a zero row nor two equal rows, so these four lengths must be 1, 2, 3, 4. In particular, $t_1 > L_2$, or $\nu_1 > \lambda_2 - 1$. The difference $\nu_1 - \lambda_2$ is integral because the lattices agree. Hence $\nu_1 \geq \lambda_2$. $\square$

For each u , choose a rotation carrying e_9 to $\hat{u}$. It conjugates K to K_u ; transport the torus and positive roots by this conjugation. Lemma 3.1 then holds for K_u in the transported weight coordinates.

Lemma 3.2 (Types on a transverse slice). *Let $\Psi \in \mathcal{C} \cap \mathcal{H}_\lambda$. For the fixed parameters in (3.1), the function*

$$\Phi(z) = \Psi(x(u, s, z))$$

belongs, as a vector in $L^2((u^\perp)^2; \mathcal{F})$, only to K_u types whose highest weights satisfy $\nu_1 \geq \lambda_2$. The K_u action here rotates z and acts by U on $\mathcal{F}$.

Proof. To restrict at fixed parameters, we first use the finite-dimensional span of the smooth translates of Ψ . The isotypic representation has the form $\mathcal{H}_\lambda \simeq V_\lambda \hat{\otimes} \mathcal{M}_\lambda$, where $\mathcal{M}_\lambda$ is a multiplicity Hilbert space. Writing $d = \dim V_\lambda$, any individual vector has an expansion $\sum_{j=1}^d e_j \otimes m_j$. Its span under G is contained in $V_\lambda \otimes \text{span}\{m_1, \dots, m_d\}$ and is therefore finite dimensional. Apply this to Ψ , and denote its translate span by W_Ψ .

Every vector in W_Ψ has a smooth compactly supported representative, because it is a finite linear combination of translates of Ψ . An equality between these representatives in L^2 holds pointwise by continuity. Thus restriction at the fixed u, s_b is a well-defined linear map from W_Ψ to $C_c^\infty((u^\perp)^2; \mathcal{F})$. It intertwines K_u , which fixes u and the longitudinal coordinates s_b . Its image can have only types occurring in $W_\Psi|_{K_u}$. Lemma 3.1 gives the claimed bound. The K_u action on the fiber L^2 space is unitary, so distinct isotypic subspaces there are orthogonal. $\square$

Choose $M \geq 0$ so that the first coordinate of every K weight of $\mathcal{F}$ is at most M . Such an M exists because $\mathcal{F}$ is finite dimensional. The same bound holds for K_u in the transported coordinates.

The factor $1/4$ in the next estimate uses half of the kinetic term in (2.1); the remaining quarter will provide the local H^1 bound needed for compactness.

Lemma 3.3 (Transverse oscillator bound). *Let $m \geq 0$ be an integer and suppose*

$$(3.5) \quad \lambda_2 > m + M.$$

For every slice Φ in Lemma 3.2,

$$(3.6) \quad \int_{(u^\perp)^2} \left(\frac{1}{4} |\nabla_z \Phi|^2 + |u|^2 |z|^2 \|\Phi\|^2 \right) dz \geq (m+9)|u| \int_{(u^\perp)^2} \|\Phi\|^2 dz.$$

Proof. For $r = |u| > 0$, the oscillator

$$h_r = -\frac{1}{4}\Delta_z + r^2|z|^2$$

has energies $r(n+8)$, $n = 0, 1, \dots$. Indeed, the change of variables $\xi = \sqrt{2r}z$ gives $h_r = \frac{r}{2}(-\Delta_\xi + |\xi|^2)$ in sixteen dimensions. Each vector at level n is an invariant Gaussian times a polynomial of total degree at most n . Tensor this level with $\mathcal{F}$.

The transverse coordinates are two copies of the vector representation of K_u , whose weights are $\pm\epsilon_1, \dots, \pm\epsilon_4$. Every weight in a polynomial of degree at most n has first coordinate at most n , and every weight in the level tensored with $\mathcal{F}$ consequently has first coordinate at most $n+M$. In particular, the first coordinate of the highest weight of any constituent is at most $n+M$.

Let Π_n be the projection onto level n tensored with $\mathcal{F}$. It commutes with K_u and hence with every K_u -isotypic projection on the fiber L^2 space. By Lemma 3.2, $\Pi_n\Phi$ can therefore contain only types with $\nu_1 \geq \lambda_2$. Since it lies in level n tensored with $\mathcal{F}$, its types also satisfy $\nu_1 \leq n+M$. For $n \leq m$ these bounds are incompatible by (3.5), so $\Pi_n\Phi = 0$. Expanding in the complete Hermite basis, the least remaining energy is $r(m+9)$, which proves (3.6). $\square$

Proposition 3.4 (Confinement in a rotation sector). *Let C be as in (2.3), and choose an integer $m \geq 0$ such that $c_m = (m+9)/3 - C > 0$. If $\lambda_2 > m+M$, every $\Psi \in \text{Dom}(q) \cap \mathcal{H}_\lambda$ belongs to $H^1(\mathbb{R}^{27}; \mathcal{F})$, satisfies $|x|^{1/2}\Psi \in L^2(\mathbb{R}^{27}; \mathcal{F})$, and obeys*

$$(3.7) \quad q(\Psi) \geq \frac{1}{4}\|\nabla\Psi\|_2^2 + c_m \int_{\mathbb{R}^{27}} |x| \|\Psi(x)\|^2 dx.$$

The operator H_λ has compact resolvent and $\ker H_\lambda = \{0\}$.

Proof. First take $\Psi \in \mathcal{C} \cap \mathcal{H}_\lambda$ and fix a color a . For fixed u , the decomposition of the two other color variables into s_b, z_b is an orthogonal linear change of variables with unit Jacobian. The transverse gradient uses only derivatives in those two variables, so $|\nabla_z \Phi|^2 \leq |\nabla \Psi|^2$. No derivative of the decomposition with respect to u is taken. Fubini's theorem, using local orthogonal frames for $u^\perp$, therefore applies to (3.6); the set $u = 0$ has measure zero. Together with (3.2), it gives

$$(3.8) \quad \frac{1}{4}\|\nabla\Psi\|_2^2 + \int V(x) \|\Psi(x)\|^2 dx \geq (m+9) \int |x^a| \|\Psi(x)\|^2 dx.$$

Average (3.8) over the three colors; its left side is unchanged. Splitting the kinetic term in (2.1) into two quarters and using (2.3) then gives

$$q(\Psi) \geq \frac{1}{4}\|\nabla\Psi\|_2^2 + \int_{\mathbb{R}^{27}} \left(\frac{m+9}{3} \sum_a |x^a| - C|x| \right) \|\Psi(x)\|^2 dx.$$

Since $\sum_a |x^a| \geq |x|$, this proves (3.7) on the core.

We now pass to the restricted form domain. By Lemma 2.2, any vector in that domain is a form-norm limit of vectors in $\mathcal{C} \cap \mathcal{H}_\lambda$. Applied to differences of these vectors, (3.7) shows convergence of their gradients in L^2 and of their products with $|x|^{1/2}$ in L^2 . The gradient and multiplication by $|x|^{1/2}$ are closed operators. Their limits are therefore the gradient and weighted product of the form limit. Passing to the limit proves (3.7) on the full restricted form domain.

Its form-norm unit ball is bounded in $H^1(\mathbb{R}^{27}; \mathcal{F})$. Moreover, for $R > 0$, (3.7) gives the uniform tail estimate

$$\int_{|x|>R} \|\Psi(x)\|^2 dx \leq \frac{q(\Psi)}{c_m R}.$$

Rellich compactness on bounded balls applies componentwise in the finite-dimensional space $\mathcal{F}$. A diagonal subsequence and the tail estimate give compactness in $L^2(\mathbb{R}^{27}; \mathcal{F})$. Hence the inclusion of the restricted form domain into $\mathcal{H}_\lambda$ is compact. The resolvent $(H_\lambda + 1)^{-1}$ maps the Hilbert space boundedly into that form domain, so it is compact.

If $\Psi \in \ker H_\lambda$, then $q(\Psi) = 0$. The positive weighted term in (3.7) vanishes, forcing $\Psi = 0$ because $|x| > 0$ almost everywhere. This proves the kernel assertion. $\square$

4. PHYSICAL ROTATION SECTORS AND POSITIVE EIGENVALUES

It remains to produce an infinite-dimensional physical sector meeting (3.5). We first obtain a gauge-invariant fermionic vector, then multiply a rotation highest vector by gauge-invariant spatial polynomials.

Lemma 4.1. *The subspace $\mathcal{F}^{\text{SU}(2)}$ is nonzero and invariant under G .*

Proof. For $1 \leq j \leq 8$ and $1 \leq a \leq 3$, put

$$c_j^a = \frac{\theta_{2j-1}^a + i\theta_{2j}^a}{\sqrt{2}}.$$

These twenty-four operators and their adjoints satisfy the canonical anticommutation relations $\{c_j^a, c_k^b\} = 0$ and $\{c_j^a, (c_k^b)^*\} = \delta_{jk}\delta_{ab}$. The occupation operators $N_j^a = (c_j^a)^*c_j^a$ are commuting projections. Start with a nonzero joint eigenvector. Applying an occupied mode's annihilator removes that occupation and preserves all others. The result is nonzero, since $\|c_j^a v\|^2 = \langle v, N_j^a v \rangle = \|v\|^2$. Repeating this gives a nonzero vector annihilated by every c_j^a . For each subset of modes, apply its creation operators in a fixed order. The resulting vectors are nonzero and have distinct occupation patterns, hence are orthogonal. Their span is invariant under every creation and annihilation operator, so irreducibility makes it all of $\mathcal{F}$. The common kernel of the annihilators is therefore exactly the original line.

Gauge transformations mix the color indices and preserve the span of the annihilators for each j . They therefore preserve the common kernel line. Their action on that line is a continuous character of $\text{SU}(2)$ and is trivial. Thus $\mathcal{F}^{\text{SU}(2)} \neq 0$. Its G invariance follows from the commutation in Lemma 2.2. $\square$

Lemma 4.2. *There are G types λ with arbitrarily large λ_2 for which $\mathcal{C} \cap \mathcal{H}_\lambda$ is infinite dimensional.*

Proof. Choose a nonzero highest vector v of some irreducible G type μ in the finite-dimensional representation $\mathcal{F}^{\text{SU}(2)}$.

We construct a gauge-invariant polynomial that raises the first two highest-weight coordinates. In the complexified vector representation, set

$$w_1 = e_1 + ie_2, \quad w_2 = e_3 + ie_4,$$

choosing the torus convention in which their weights are ϵ_1, ϵ_2 . Use the complex bilinear extension of the Euclidean dot product, and put

$$\zeta_a = w_1 \cdot x^a, \quad \eta_a = w_2 \cdot x^a, \quad P(x) = \sum_{a < b} (\zeta_a \eta_b - \zeta_b \eta_a)^2.$$

The identity

$$P = (\zeta \cdot \zeta)(\eta \cdot \eta) - (\zeta \cdot \eta)^2$$

uses bilinear dot products on the color space $\mathbb{C}^3$. The real orthogonal adjoint action of $\text{SU}(2)$ preserves these products, so P is gauge invariant. At $x = t(e_1, e_3, 0)$ with $t > 0$, one has $P(x) = t^4$ and $|x| = \sqrt{2}t$. Thus P is nonzero on every nonempty radial shell.

For the action on spatial polynomials by pullback,

$$(w \cdot R_g^{-1}x) = (R_g w) \cdot x.$$

Thus $w \mapsto (w \cdot x^a)$ intertwines the complexified vector representation with linear polynomials, with the same weight sign. The analogous map on exterior squares sends $w_1 \wedge w_2$ to $\zeta_a \eta_b - \zeta_b \eta_a$.

The bivector $w_1 \wedge w_2$ is highest of weight $\epsilon_1 + \epsilon_2$: a positive-root operator either kills both factors or sends w_2 to a multiple of w_1 , and in the latter case its action on the wedge also vanishes. Hence each minor is highest of that weight. Root operators act as derivations on polynomials, so P is highest of weight $2(\epsilon_1 + \epsilon_2)$. Its nonzero power P^k is highest of weight $2k(\epsilon_1 + \epsilon_2)$.

Let $\chi \in C_c^\infty((0, \infty))$ be nonzero. The function

$$\Psi_\chi(x) = \chi(|x|)P(x)^k v$$

is in $\mathcal{C}$: it is smooth because its support avoids the origin, and its factors are gauge invariant. The preceding ray shows that $\Psi_\chi \neq 0$. Multiplication by the radial function is G equivariant. The translates of Ψ_χ lie in the finite-dimensional representation obtained from homogeneous polynomials of degree $4k$ tensored with $\mathcal{F}$. The vector is highest of the dominant weight

$$\lambda = 2k(\epsilon_1 + \epsilon_2) + \mu = (2k + \mu_1, 2k + \mu_2, \mu_3, \mu_4).$$

In a decomposition of that finite representation into irreducibles, each nonzero component of this vector is a weight- λ vector killed by all positive-root operators. It must be a highest vector, so its summand has type λ . Therefore $\Psi_\chi \in \mathcal{H}_\lambda$.

Choosing χ with pairwise disjoint radial supports gives infinitely many linearly independent vectors in the same sector. Finally, $\lambda_2 = 2k + \mu_2$ tends to infinity with k . $\square$

Proof of Theorem 1.1. Choose an integer $m \geq 0$ with $c_m = (m + 9)/3 - C > 0$. Lemma 4.2 supplies a type λ such that $\lambda_2 > m + M$ and $\mathcal{H}_\lambda$ is infinite dimensional. By Proposition 3.4, H_λ has compact resolvent and no zero eigenvector.

Since H_λ is nonnegative with compact resolvent on an infinite-dimensional Hilbert space, the compact-resolvent spectral theorem gives an orthonormal eigenbasis (Ψ_j) with eigenvalues $E_j \rightarrow \infty$. Its trivial kernel makes every E_j positive. By Lemma 2.2, H_λ is the part of H on a reducing subspace. These vectors therefore belong to $\text{Dom}(H)$ and satisfy $H\Psi_j = E_j\Psi_j$ in the full physical Hilbert space. $\square$

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